Experimental Stress Analysis

UNIT – III

STRAIN ROSETTES AND NON-DESTRUCTIVE TESTING OF CONCRETE

Course 23A01702c · Professional Elective-IV · JNTUA R23

Syllabus (this unit):

Introduction – The Three-Element Rectangular Rosette – The Delta Rosette – Corrections for Transverse Strain Gauge – Ultrasonic Pulse Velocity Method – Application to Concrete – Hammer Test – Application to Concrete.

Handwritten-style study notes · B.Tech Civil Engineering – IV Year

1. Introduction – Why Strain Rosettes?

- A single strain gauge measures strain only along its own axis.
- At a point on a free surface (plane stress) there are three unknowns:

$$\varepsilon_x, \varepsilon_y, \gamma_{xy} \text{ or equivalently } \varepsilon_1, \varepsilon_2, \theta \text{ (principal strains and direction)}$$

- Hence three independent strain readings in three different directions are needed. A group of three (or four) gauges at one point is called a strain rosette.
- If the principal directions are known (simply supported beam, bar in tension, pressure vessel) – two gauges (0° and 90°) or even one gauge is enough.
- If the directions are unknown (joints, plates with holes, bracket, around openings) – a rosette is a must.

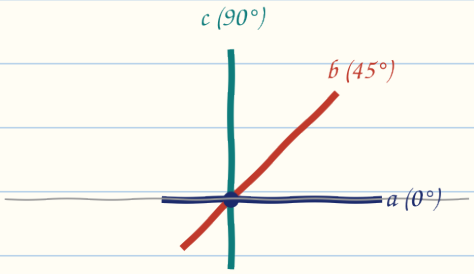
Strain at any angle θ (transformation equation)

$$\varepsilon_\theta = (\varepsilon_x + \varepsilon_y)/2 + (\varepsilon_x - \varepsilon_y)/2 \cdot \cos 2\theta + (\gamma_{xy}/2) \cdot \sin 2\theta$$

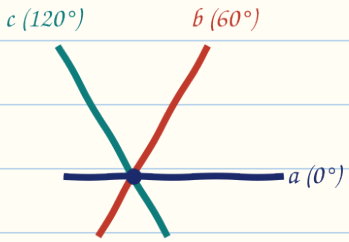
- Writing this for three gauge angles $\theta_a, \theta_b, \theta_c$ gives three equations for the three unknowns.

Common rosette types

- Rectangular (0°-45°-90°), Delta (0°-60°-120°), Tee-delta, and four-element 0-45-90-135 (extra gauge gives a check $\varepsilon_a + \varepsilon_c = \varepsilon_b + \varepsilon_d$).
- Built as planar (all grids on one foil – cheap, small) or stacked (grids one above another – measure at one exact point; poorer heat dissipation).



Rectangular rosette (0°, 45°, 90°)



Delta rosette (0°, 60°, 120°)

Fig 3.1 Rectangular and delta rosettes

2. Three-Element Rectangular Rosette

- Gauges a, b, c at 0°, 45° and 90° to the reference x-axis; readings $\varepsilon_a, \varepsilon_b, \varepsilon_c$.
- Substituting $\theta = 0°, 45°, 90°$ in the transformation equation:

$$\varepsilon_a = \varepsilon_x \quad \varepsilon_c = \varepsilon_y \quad \varepsilon_b = (\varepsilon_x + \varepsilon_y)/2 + \gamma_{xy}/2$$
$$\therefore \gamma_{xy} = 2\varepsilon_b - \varepsilon_a - \varepsilon_c$$

Principal strains and direction

$$\varepsilon_{1,2} = (\varepsilon_a + \varepsilon_c)/2 \pm (1/\sqrt{2})\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2]}$$
$$\tan 2\theta_n = (2\varepsilon_b - \varepsilon_a - \varepsilon_c)/(\varepsilon_a - \varepsilon_c) \quad (\theta_n \text{ measured from gauge a})$$
$$\gamma_{max} = \varepsilon_1 - \varepsilon_2 = \sqrt{2}\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2]}$$

Principal stresses (Hooke's law, plane stress)

$$\sigma_{1,2} = (E/2)\{(\varepsilon_a + \varepsilon_c)/(1-\nu) \pm [\sqrt{2}/(1+\nu)]\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2]}\}$$
$$\tau_{max} = (\sigma_1 - \sigma_2)/2 = E(\varepsilon_1 - \varepsilon_2)/[2(1+\nu)]$$

- Mohr's circle of strain gives the same results graphically: plot $(\varepsilon_x, \gamma_{xy}/2)$ and $(\varepsilon_y, -\gamma_{xy}/2)$; centre at $(\varepsilon_x + \varepsilon_y)/2$.

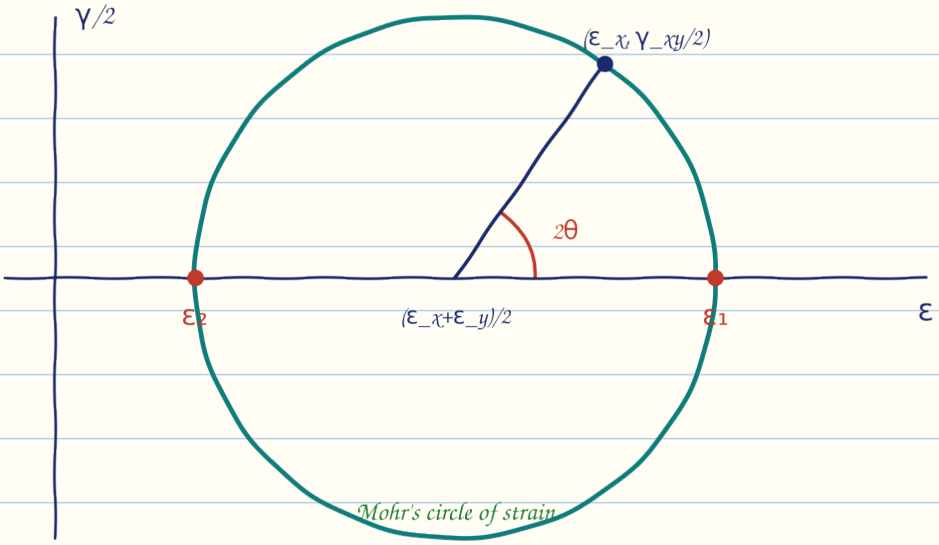


Fig 3.2 Mohr's circle of strain for rosette readings

Steps to construct Mohr's circle of strain from rosette readings

- First find $\varepsilon_x = \varepsilon_a$, $\varepsilon_y = \varepsilon_c$ and $\gamma_{xy} = 2\varepsilon_b - \varepsilon_a - \varepsilon_c$.
- Draw axes: ε horizontal, $\gamma/2$ vertical (clockwise shear taken positive downward by the usual convention).
- Plot points X $(\varepsilon_x, \gamma_{xy}/2)$ and Y $(\varepsilon_y, -\gamma_{xy}/2)$; join them – the line cuts the ε-axis at the centre $C = (\varepsilon_x + \varepsilon_y)/2$.

- Draw the circle with centre C and radius $R = XC$. Its intercepts on the ε-axis give ε_1 (right) and ε_2 (left); top of circle gives $\gamma_{max}/2 = R$.
- The angle XC to the ε-axis is $2\theta_p$; the principal axis is rotated by θ_p from gauge a (in the same sense as the circle angle).

Check with a known field (uniaxial tension)

- Let the rosette lie along a bar with uniaxial stress σ in direction a: $\varepsilon_a = \varepsilon = \sigma/E$, $\varepsilon_c = -\nu\varepsilon$, and since $\gamma_{xy} = 0$, $\varepsilon_b = (\varepsilon_a + \varepsilon_c)/2 = \varepsilon(1 - \nu)/2$.

$$\varepsilon_1 = \varepsilon(1 - \nu)/2 + (1/\sqrt{2})\,\varepsilon\,(1 + \nu)/2 \cdot \sqrt{2} = \varepsilon(1 - \nu)/2 + \varepsilon(1 + \nu)/2 = \varepsilon \text{ and } \theta_p = 0°$$

- So the formulae return the true strain and the correct axis – a useful check on a new rosette installation.

Practical points for rosette installation

- Align the a-gauge with the reference axis (scribe line) within ±1°; an angular error of 1° gives a noticeable error in θ_p and shear.
- Connect each element as a quarter bridge with its own dummy (or a common dummy) and use the same lead length for all three.
- Keep the zero reading before loading and record all three channels at the same load step.

Solved example 1

- Rectangular rosette on steel ($E = 200$ GPa, $\nu = 0.3$): $\varepsilon_a = 800$, $\varepsilon_b = 400$, $\varepsilon_c = -200\ \mu\varepsilon$.

$$\text{Centre} = (800 - 200)/2 = 300\ \mu\varepsilon;\ (\varepsilon_a - \varepsilon_b) = 400,\ (\varepsilon_b - \varepsilon_c) = 600$$

$$\text{radius} = (1/\sqrt{2})\sqrt{(400^2 + 600^2)} = 509.9\ \mu\varepsilon$$

$$\varepsilon_1 = 809.9\ \mu\varepsilon,\ \varepsilon_2 = -209.9\ \mu\varepsilon$$

$$\tan 2\theta_p = (800 - 800 + 200)/(800 + 200) = 0.2 \rightarrow 2\theta_p = 11.3°,\ \theta_p = 5.65° \text{ from gauge a}$$

$$\sigma_1 = E/(1 - \nu^2)\,(\varepsilon_1 + \nu\,\varepsilon_2) = (200\times10^3/0.91)(809.9 - 63.0)\times10^{-6} = 164.2\ \text{MPa}$$

$$\sigma_2 = (200\times10^3/0.91)(-209.9 + 243.0)\times10^{-6} = 7.3\ \text{MPa}$$

3. The Delta Rosette (0°-60°-120°)

- Gauges a, b, c at 0°, 60°, 120° (equiangular – 60° apart).

$$\varepsilon_a = \varepsilon_x \quad \varepsilon_b = \varepsilon_x/4 + 3\varepsilon_y/4 + (\sqrt{3}/4)\gamma_{xy} \quad \varepsilon_c = \varepsilon_x/4 + 3\varepsilon_y/4 - (\sqrt{3}/4)\gamma_{xy}$$

- Solving:

$$\varepsilon_x = \varepsilon_a \quad \varepsilon_y = [2(\varepsilon_b + \varepsilon_c) - \varepsilon_a] / 3 \quad \gamma_{xy} = 2(\varepsilon_b - \varepsilon_c)/\sqrt{3}$$

Principal strains, directions and stresses

$$\varepsilon_{1,2} = (\varepsilon_a + \varepsilon_b + \varepsilon_c)/3 \pm (\sqrt{2}/3)\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2 + (\varepsilon_c - \varepsilon_a)^2]}$$

$$\tan 2\theta_n = \sqrt{3}(\varepsilon_b - \varepsilon_c) / (2\varepsilon_a - \varepsilon_b - \varepsilon_c)$$

$$\sigma_{1,2} = E\{(\varepsilon_a + \varepsilon_b + \varepsilon_c)/[3(1-\nu)] \pm [\sqrt{2}/(3(1+\nu))]\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2 + (\varepsilon_c - \varepsilon_a)^2]}\}$$

Solved example 2

- Delta rosette on steel ($E = 200$ GPa, $\nu = 0.3$): $\varepsilon_a = 600$, $\varepsilon_b = 300$, $\varepsilon_c = -100$ $\mu\varepsilon$.

$(\varepsilon_a + \varepsilon_b + \varepsilon_c)/3 = 266.7\ \mu\varepsilon$; $\Sigma(\text{difference})^2 = 300^2 + 400^2 + 700^2 = 740\,000 \rightarrow \sqrt{} = 860.2$

radius $= (\sqrt{2}/3)(860.2) = 405.5\ \mu\varepsilon \rightarrow \varepsilon_1 = 672.2\ \mu\varepsilon$, $\varepsilon_2 = -138.8\ \mu\varepsilon$

$\tan 2\theta_n = \sqrt{3}(400)/(1200 - 300 + 100) = 0.693 \rightarrow \theta_n = 17.4°$ from gauge a

$\sigma_1 = (200\times10^3/0.91)(672.2 - 41.6)\times10^{-6} = 138.6$ MPa; $\sigma_2 = (200\times10^3/0.91)(-138.8 + 201.7)\times10^{-6} = 13.8$ MPa

Rectangular vs delta

- Rectangular: simpler formulae, ε_x and ε_y read directly; the most used.
- Delta: all gauges equally spaced (60°), better uniform accuracy in every direction and more compact; formulae slightly longer.
- Both give identical results if the readings correspond to the same strain field.

4. Corrections for Transverse Strain (Transverse Sensitivity)

- A strain gauge grid has end loops (turn-around) and a finite width, so it responds slightly to the strain perpendicular to its axis.

$$\Delta R/R = S_a(\varepsilon_a + K_t\varepsilon_t) \quad K_t = S_t/S_a \text{ (transverse sensitivity factor)}$$

- S_a = axial sensitivity, S_t = transverse sensitivity; K_t is typically 0 – 5 % (foil gauges ≈ 0 – 2 %, wire-wrap gauges higher).
- Manufacturer's gauge factor S_g is measured on a beam of Poisson's ratio $\nu_0 = 0.285$, where $\varepsilon_t = -\nu_0\varepsilon_a$, so

$$S_g = S_a (1 - \nu_o K_t)$$

- Indicated strain (using S_g): $\varepsilon' = (\varepsilon_a + K_t \varepsilon_t) / (1 - \nu_o K_t)$. True strains for a 0°/90° (two-gauge) arrangement:

$$\varepsilon_a = [(1 - \nu_o K_t)/(1 - K_t^2)] (\varepsilon'_a - K_t \varepsilon'_c)$$
$$\varepsilon_c = [(1 - \nu_o K_t)/(1 - K_t^2)] (\varepsilon'_c - K_t \varepsilon'_a)$$

- For the 45° gauge of a rectangular rosette (transverse strain = $\varepsilon_a + \varepsilon_c - \varepsilon_b$):

$$\varepsilon_b = [(1 - \nu_o K_t)/(1 - K_t)] [\varepsilon'_b - K_t (\varepsilon'_a + \varepsilon'_c)/(1 + K_t)]$$

Solved example 3

- $K_t = 0.02$, $\nu_o = 0.285$; indicated $\varepsilon'_a = 1000\ \mu\varepsilon$ and $\varepsilon'_c = -300\ \mu\varepsilon$.

$$\text{factor} = (1 - 0.0057)/(1 - 0.0004) = 0.9947$$

$\varepsilon_a = 0.9947\ (1000 + 0.02 \times 300) = 1000.7\ \mu\varepsilon$ $\varepsilon_c = 0.9947\ (-300 - 0.02 \times 1000) = -318.3\ \mu\varepsilon$

- The error in the smaller strain ε_c is about 6 %, while ε_a has a negligible error. So correct whenever the transverse strain is large compared with the axial strain, or K_t is above 1 %.

Other corrections in rosette work

- Temperature (use self-temperature-compensated gauges and a dummy gauge), lead-wire resistance, non-linearity of bridge at large strains, finite size of the rosette (stacked rosettes reduce this), reinforcing effect of gauge.

5. Non-Destructive Testing (NDT) of Concrete – Introduction

- NDT finds strength, quality, cracks, voids and deterioration of concrete without damaging the structure.
- Why needed: doubtful cube results, old structures, fire/earthquake damage, quality control during construction, deciding formwork removal time.
- Main methods: rebound (Schmidt) hammer, ultrasonic pulse velocity (UPV), penetration resistance (Windsor probe), pull-out test, cover meter, half-cell potential (corrosion), radiography, core cutting (partially destructive).
- Standards: IS 13311 (Part 1) – UPV; IS 13311 (Part 2) – Rebound hammer; IS 516 (Part 5).

6. Ultrasonic Pulse Velocity (UPV) Method

6.1 Principle

- An ultrasonic pulse (frequency 24 – 150 kHz, usually 54 kHz) is sent through concrete from a transmitting transducer; a receiving transducer records the transit time T.

  Pulse velocity $V = L / T$ (L = path length, T = transit time)

- Sound travels faster in dense, good concrete than in porous, cracked concrete, so velocity indicates quality.
- The pulse velocity depends on the elastic constants and density:

  $$V = \sqrt{\{E(1-\mu)/[\rho(1+\mu)(1-2\mu)]\}} \rightarrow E_d = \rho V^2 (1+\mu)(1-2\mu)/(1-\mu)$$

6.2 Equipment

- Pulse generator, two piezo-electric transducers (T and R), amplifier, time-interval measuring circuit (resolution 0.1 μs), display; calibration (reference) bar of known time; couplant (grease / jelly) for good acoustic contact.

6.3 Arrangement of transducers

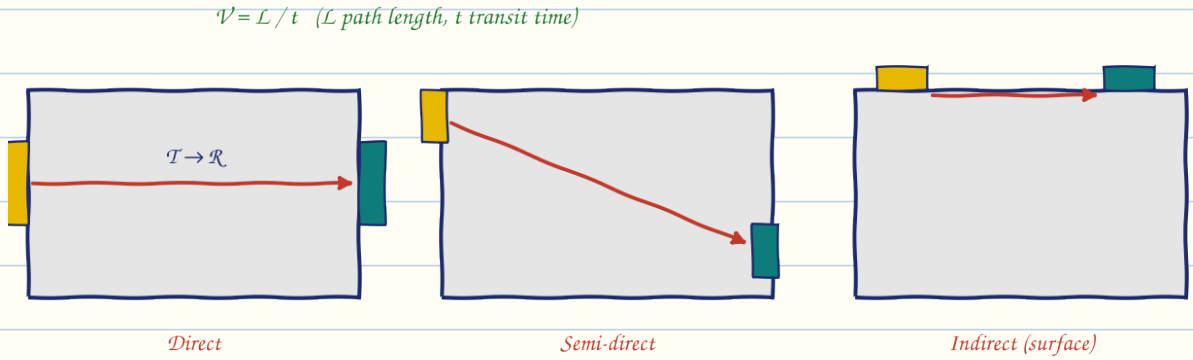


Fig 3.3 Direct, semi-direct and indirect transmission

- Direct (cross) transmission – transducers on opposite faces; most reliable, maximum pulse energy – preferred.
- Semi-direct – on adjacent faces; less sensitive, path length is measured centre-to-centre.
- Indirect (surface) – both transducers on the same face; weakest signal, velocity is 5–20 % lower; find V from the slope of the $T – L$ graph for several distances; used where only one face is accessible or to detect surface cracks.

- Minimum path length: 100 mm for aggregate ≤ 20 mm and 150 mm for aggregate ≤ 40 mm.

6.4 Procedure

- Set the reference-bar time; smooth the surface; apply couplant; hold the transducers firmly; measure L accurately; read T (take 3 readings); compute $V = L/T$.

6.5 Quality of concrete (IS 13311 Part 1)

- V above 4.5 km/s → Excellent; 3.5 – 4.5 → Good; 3.0 – 3.5 → Medium (doubtful); below 3.0 → Poor.

6.6 Factors affecting pulse velocity

- Moisture (wet concrete – higher V), aggregate type and quantity (more aggregate – higher V), w/c ratio (lower – higher V), age (increases with age), reinforcement (velocity along steel is higher – keep path away from bars, apply correction), cracks and voids (longer path – lower V), path length (> 100 mm), temperature (10–30 °C – negligible), stress level and transducer contact.

6.7 Applications of UPV to concrete

- Homogeneity (uniformity) check at many points; detect honeycombing, voids, cracks.
- Depth of surface-opening crack – transducers placed at equal distances x on either side of crack:

$$h = x\sqrt{[\,(T_c / T_s)^2 - 1\,]}$$ (T_c = time across crack, T_s = time over $2x$ on sound concrete)

- Strength estimation – V–f_c correlation graph made from cores / cubes of the same concrete; accuracy ± 15–25 %.
- Dynamic modulus of elasticity E_d; monitoring early-age setting, fire damage, freeze-thaw, sulphate attack, alkali-aggregate reaction.

Strength correlation with UPV (approximate guide)

- V = 3.5 km/s → about 15 – 20 N/mm²; V = 4.0 km/s → 25 – 35 N/mm²; V = 4.5 km/s → 40 – 50 N/mm². These vary with aggregate and mix, so a site-specific curve from cores or cubes must be drawn (log f_c vs V is usually a

straight line).

- *Procedure for the calibration curve:* cast cubes of different strength (change w/c); measure V through the cube before crushing, crush to get f_c, plot f_c against V and fit a curve.
- *Example:* $L = 300$ mm, $T = 71.4$ μs → $V = 300/71.4 = 4.2$ km/s (good). With $\rho = 2400$ kg/m³ and $\mu = 0.2$:

$$E_d = 2400 \times 4200^2 \times (1.2 \times 0.6/0.8) = 3.8\times10^{10}\ \text{Pa} = 38\ \text{GPa}$$

- *Crack-depth example:* $x = 150$ mm, $T_s = 70$ μs, $T_c = 98$ μs → $h = 150\sqrt{(1.4^2 - 1)} = 147$ mm.
- *Limitations:* needs calibration for strength, not directly strength; influenced by moisture and steel; cost of equipment.

7. Rebound (Schmidt) Hammer Test

7.1 Principle

- A *spring-loaded mass* strikes a steel plunger resting on the concrete surface. The *rebound distance* of the mass depends on the *surface hardness*, which is related to the *strength* of concrete. Rebound reading is the *rebound number* R (scale 10–100).
- Impact energy: 2.25 N·m (type N, ordinary concrete); 0.735 N·m for small members (type L).

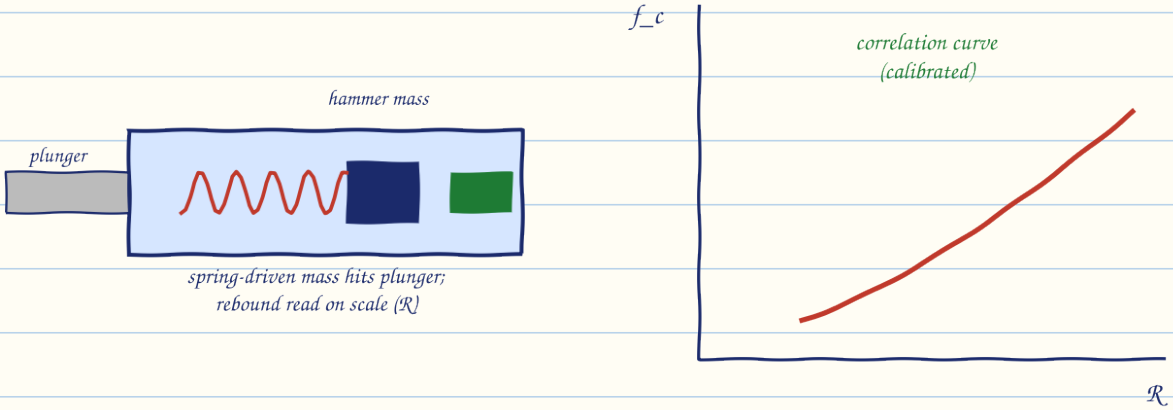


Fig 3.4 Rebound hammer and its correlation curve

7.2 Procedure (IS 13311 Part 2)

- Calibrate on the steel anvil supplied ($R \approx 80 \pm 2$).
- Choose a smooth, clean, dry, surface (grind rough patches; avoid honeycomb, rebar, edges within 20 mm).
- Hold the hammer perpendicular to the surface; push the plunger slowly; after impact note R.
- Take at least 9 (usually 12) readings on a 300 mm × 300 mm area, points ≥ 20 mm apart, discard readings deviating more than ±6 from the average and re-average.
- Apply the orientation correction (upward / downward / vertical) and use the correlation curve to obtain compressive strength.

7.3 Typical correlation (cubes, horizontal impact)

- $R \approx 20 \rightarrow$ about 10 – 15 N/mm^2; $R \approx 30 \rightarrow 20 – 25$ N/mm^2; $R \approx 40 \rightarrow 35 – 40$ N/mm^2; $R \approx 50 \rightarrow 55 – 60$ N/mm^2 (approximate – use the manufacturer's calibration chart).

7.4 Factors affecting rebound number

- Smoothness of surface; moisture (wet surface lowers R by up to 20 %); carbonation (raises R, remove the skin); age and curing; type of cement, aggregate and mix; size and stiffness of the member (thin members absorb impact); stress state, temperature, direction of impact.

7.5 Application to concrete

- Uniformity of concrete in a structure; locating weak zones; comparing the quality of different members; checking early strength (formwork removal); assessing old buildings; supplementary to core tests.
- Advantages: quick, cheap, light, simple, many readings in short time.
- Limitations: measures surface hardness (top 30 mm only), not true strength; accuracy ± 25 %; results depend on moisture, carbonation, aggregate; cannot replace cube or core tests.
- Combined method (SonReb): use R and V together in a multiple-regression formula; error reduces to about ± 10 %.

Solved example 4 – rebound readings

- Twelve readings at a point: 34, 36, 35, 33, 38, 34, 35, 36, 35, 27, 34, 35.

  Average of all = 412/12 = 34.3; 27 differs by 7.3 > 6 → reject

  Average of the remaining 11 = 385/11 = 35.0; all others lie within ±6 of 35 → $\mathcal{R}$ = 35

- From the calibration chart $\mathcal{R}$ = 35 → compressive strength ≈ 28 – 32 N/mm² (M30-type concrete).
- If the hammer pointed downward, the actual $\mathcal{R}$ would be corrected by about −3 (downward impact gives higher $\mathcal{R}$); for upward impact add about +3 (values from the chart supplied with the instrument).

Comparison

- UPV – looks through the depth, finds voids and cracks, quality grading, more costly. Rebound hammer – surface only, strength indicator, very cheap and quick. Both together give a reliable assessment.

8. Quick Revision – Unit III

- Rosette = 3 gauges at a point → $\varepsilon_1, \varepsilon_2, \theta$ (plane stress).

  Rectangular: $\varepsilon_{1,2} = (\varepsilon_a + \varepsilon_c)/2 \pm (1/\sqrt{2})\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2]}$, $\tan 2\theta = (2\varepsilon_b - \varepsilon_a - \varepsilon_c)/(\varepsilon_a - \varepsilon_c)$

  Delta: $\varepsilon_{1,2} = (\varepsilon_a + \varepsilon_b + \varepsilon_c)/3 \pm (\sqrt{2}/3)\sqrt{[\Sigma(\text{differences})^2]}$, $\tan 2\theta = \sqrt{3}(\varepsilon_b - \varepsilon_c)/(2\varepsilon_a - \varepsilon_b - \varepsilon_c)$

- Transverse sensitivity: $\varepsilon_a = [(1 - \nu_0 K_t)/(1 - K_t^2)](\varepsilon'_a - K_t \varepsilon'_c)$.
- UPV: $V = L/T$; > 4.5 excellent, 3.5–4.5 good, 3–3.5 medium, < 3 poor.
- Rebound hammer: $\mathcal{R}$ number → strength by chart; surface hardness; ± 25 %.

Important questions

- What is a strain rosette? Why are three gauges required? Derive the principal strains for a rectangular rosette.
- Derive the expressions for principal strains and stresses for a delta rosette.
- What is transverse sensitivity? Derive the correction for a 0°/90° gauge combination.
- Explain the principle, procedure and applications of the UPV method for concrete.

- Describe the rebound hammer test. List the factors affecting the rebound number.
- Compare the UPV and rebound hammer methods of NDT.